Midpoint Reset: A Full-Round Poseidon Collision from an Adaptively Chosen MDS Matrix

Sunghyeon Jo[1,2]

[1] Georgia Institute of Technology
[2] QED Audit
sjo65@gatech.edu

Abstract. We give an explicit compression collision for all 28 rounds of the KoalaBear Poseidon instance with parameters $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$, in the setting where the round constants are fixed before the MDS linear layer is chosen. The main problem is to construct a single linear layer that simultaneously controls two executions through both the full and partial rounds. We do this by tracking their midpoint and half-difference. In each two-round block, one prescribed image of the linear layer cancels the midpoint against the next round constant, so the following odd cubic S-box receives opposite states and resets the midpoint to zero. Two additional images are reused throughout the permutation to return the half-difference to the same one-dimensional subspace. The resulting trajectory constraints determine a linear layer, while a scalar recurrence closes the final difference under feed-forward. For the KoalaBear instance we obtain a collision in all sixteen output coordinates with an MDS matrix satisfying the prescribed linear-layer checks. The scalar construction reduces to low-degree equations and admits a family of parameter choices, so the collision is not an isolated instance. The result exposes an adaptive correlation between fixed round constants and a subsequently chosen linear layer that matrix-only checks do not capture.

Keywords: Poseidon · hash collision · MDS matrix · zero-knowledge

1 Introduction

Poseidon [5] is an arithmetization-oriented permutation based on the HADES design strategy [7]. Full rounds apply a power S-box to every state word, partial rounds apply it to one word, and each round is followed by an MDS linear layer. This structure has motivated several lines of algebraic cryptanalysis, including attacks based on reparametrization, round skipping, subspace trails, root finding, and resultants [2,6,1,9].

The Poseidon Cryptanalysis Initiative [4] considers, among other settings, a collision challenge in which the round constants are fixed while the attacker may provide an MDS matrix satisfying the prescribed linear-layer checks. The

stated rules did not explicitly prohibit choosing the matrix as a function of those constants. This raises a natural cryptanalytic question: can a matrix satisfy the prescribed matrix-level requirements yet interact with the fixed round constants in a way that creates a full collision?

We answer this question for the width-16 KoalaBear instance with eight full rounds and twenty partial rounds. The attack is based on a two-execution trajectory. Write the states at a round boundary as

$$x_r^{\pm} = m_r \pm d_r,$$

where m_r is their midpoint and d_r is their half-difference. Let u_r denote the midpoint immediately after the first S-box layer of a two-round block. We prescribe

$$Mu_r = -c^{(r+1)},$$

where $c^{(r+1)}$ is the next round constant. After the following constant addition, the two S-box inputs are exact opposites. Since the cubic S-box is odd, their midpoint is zero after that S-box layer and remains zero after the next linear layer. We call this operation a *midpoint reset*.

The difference is controlled independently. We choose a coordinate direction e that is passive in every partial round and two reusable images

$$Me = v, \qquad Mw = e.$$

The vectors are chosen so that a two-round block returns the half-difference to the line $\mathbb{F}e$ in both the full-round and partial-round parts of the permutation. Thus every two-round block contributes one new midpoint-reset constraint, while the same two difference constraints are reused throughout the permutation.

For the concrete 28-round instance, the resulting source vectors are linearly independent. Hence the prescribed images determine a unique linear layer. The scalar difference trail is chosen so that the final permutation difference is the negative of the input difference. The feed-forward compression therefore cancels the difference and yields a collision in all sixteen output coordinates. The resulting matrix is MDS and passes the Initiative's linear-layer checks against the infinitely long subspace trails considered in [8].

The main point is the trajectory itself: once the constants are fixed, the freedom to choose the linear layer can be used to program an exact nonlinear two-execution path through all rounds. MDS and subspace-trail checks constrain the matrix in isolation; they do not constrain correlations deliberately introduced between the matrix and the round constants.

The construction applies to adaptive parameter generation in which the constants are known before the linear layer is selected. The exact matrix, construction parameters, and collision are given in Appendix A.

2 Preliminaries

2.1 Notation

Let $\mathbb{F} = \mathbb{F}_p$ be a prime field of odd characteristic. Poseidon states are represented as column vectors in $\mathbb{F}^t$, with coordinates indexed $0, \ldots, t-1$. The vector e_i denotes the i-th standard basis vector.

For $v = (v_0, \ldots, v_{t-1})^T \in \mathbb{F}^t$ and an integer $k \geq 1$, write

$$v^{\circ k} = (v_0^k, \ldots, v_{t-1}^k)^T$$

for componentwise exponentiation. All linear algebra is over $\mathbb{F}$ unless stated otherwise. A matrix $M \in \mathbb{F}^{t \times t}$ is MDS if every nonempty square submatrix of M is nonsingular.

2.2 Poseidon permutation

Let $M \in \mathbb{F}^{t \times t}$ be the MDS linear layer, let α be the S-box exponent, and let $c^{(r)} \in \mathbb{F}^t$ be the round-constant vector for round r. Define the full and partial nonlinear layers by

$$S_F(z) = z^{\circ \alpha}, \qquad S_P(z) = (z_0^\alpha, z_1, \ldots, z_{t-1})^T.$$

Let $S_r = S_F$ when round r is full and $S_r = S_P$ when it is partial. The round function is

$$R_r(x) = M S_r\big(x + c^{(r)}\big). \tag{1}$$

For the concrete KoalaBear instance,

$$p = 2^{31} - 2^{24} + 1 = 2130706433, \qquad t = 16, \qquad \alpha = 3,$$
$$R_F = 8, \qquad R_P = 20.$$

Rounds 0–3 and 24–27 are full rounds, while rounds 4–23 are partial rounds. The round constants are those of the reference instance at the pinned commit specified in [4]. We write

$$\mathsf{P}_M = R_{27} \circ \cdots \circ R_0$$

for the complete permutation. Since $\alpha = 3$, both nonlinear layers are odd:

$$S_F(-z) = -S_F(z), \qquad S_P(-z) = -S_P(z). \tag{2}$$

2.3 Compression mode

The challenge accepts a 15-word message and prepends the fixed field element

$$\rho = \texttt{0xc09de4} = 12623332.$$

We work with the resulting full state $x \in \mathbb{F}^{16}$, so $x_0 = \rho$. The feed-forward compression function is

$$\mathsf{H}_M(x) = \mathsf{P}_M(x) + x. \tag{3}$$

A collision is a pair of distinct admissible states x^+, x^- satisfying $\mathsf{H}_M(x^+) = \mathsf{H}_M(x^-)$.

2.4 Midpoint and half-difference

Let x_r^+ and x_r^- be the two states entering round r. Define their midpoint and half-difference by

$$m_r = \frac{x_r^+ + x_r^-}{2}, \qquad d_r = \frac{x_r^+ - x_r^-}{2},$$

and set

$$s_r = m_r + c^{(r)}.$$

Immediately after the nonlinear layer, define

$$u_r = \frac{S_r(s_r + d_r) + S_r(s_r - d_r)}{2}, \tag{4}$$

$$q_r = \frac{S_r(s_r + d_r) - S_r(s_r - d_r)}{2}. \tag{5}$$

The following linear layer acts on these two components independently:

$$m_{r+1} = Mu_r, \qquad d_{r+1} = Mq_r. \tag{6}$$

3 The Midpoint-Reset Construction

3.1 Resetting the midpoint across two rounds

Consider consecutive rounds r and $r+1$ and prescribe

$$Mu_r = -c^{(r+1)}. \tag{7}$$

Then

$$m_{r+1} = -c^{(r+1)},$$

so after adding the next round constant the two inputs to the S-box layer are $\pm d_{r+1}$. By (2), the nonlinear layer is odd. Therefore its output midpoint is zero:

$$u_{r+1} = 0.$$

The following linear layer preserves zero, and hence

$$m_{r+2} = 0.$$

Thus one prescribed image of M resets the midpoint over the entire two-round block.

For the cubic S-box, a full-round block beginning with half-difference be can be displayed as

$$m \pm be \xrightarrow{+c,\,(\cdot)^3} u \pm \Delta e \xrightarrow{M} -c' \pm \Delta v \xrightarrow{+c'} \pm \Delta v \xrightarrow{S_F} \pm b'w \xrightarrow{M} \pm b'e,$$

where $c' = c^{(r+1)}$. The assignment $Mu = -c'$ controls the midpoint; the assignments $Me = v$ and $Mw = e$ below control the difference.

3.2 A reusable difference cycle

Let $e = e_1$, so e is supported on a coordinate that is passive in every partial round. Fix a nonzero scalar $a \in \mathbb{F}_p$ and choose $v \in \mathbb{F}_p^{16}$. Define

$$w = a^2 v^{\circ 3}, \qquad Me = v, \qquad Mw = e. \tag{8}$$

For every passive coordinate require

$$v_i \in \{a^{-1}, -a^{-1}\} \qquad (i \geq 1). \tag{9}$$

Then $a^2 v_i^3 = v_i$ for $i \geq 1$, which makes the same vector w compatible with both full and partial rounds.

Full-round block. Suppose a full-round block begins with half-difference be, and let s be the common value in coordinate 1 before the first S-box. The first cubic S-box produces half-difference

$$\Delta e, \qquad \Delta = \frac{(s+b)^3 - (s-b)^3}{2} = b^3 + 3bs^2.$$

The first matrix multiplication sends this to Δv. After the midpoint reset, the second S-box is centered at zero and maps the half-difference to

$$\Delta^3 v^{\circ 3} = \frac{\Delta^3}{a^2} w.$$

The assignment $Mw = e$ therefore returns the difference to $\mathbb{F}e$, with scalar update

$$b' = a^{-2}\left(b^3 + 3bs^2\right)^3. \tag{10}$$

Partial-round block. Suppose a centered partial-round block begins with half-difference ae. Because coordinate 1 is passive, the first S-box leaves this half-difference unchanged, and $Me = v$ sends it to av. The second partial S-box is centered at zero. Its active coordinate becomes $a^3 v_0^3$, while each passive coordinate remains av_i. By (9), this vector is exactly aw. The assignment $Mw = e$ therefore returns the half-difference to ae. By oddness, the same statement holds at amplitude $-a$. Consequently all ten partial-round blocks reuse the same two linear images without adding new difference constraints.

3.3 Assembling the linear layer

Let

$$u_0, u_2, \ldots, u_{26}$$

be the post-S-box midpoints at the first round of the fourteen two-round blocks. We impose

$$Mu_r = -c^{(r+1)} \qquad (r = 0, 2, \ldots, 26), \tag{11}$$

together with the two reusable assignments in (8).

The source vectors in these assignments are determined before M is constructed. Indeed, the reset at round r forces $m_{r+2} = 0$, so every even-round midpoint after the first is known. The difference cycle keeps $d_r = b_r e$ at each even-round boundary, with b_r determined by the scalar recurrence (10) in full-round blocks and fixed at $\pm a$ in the partial-round blocks. Hence each u_r is determined by the chosen initial midpoint, the public round constants, and the scalar trail; it does not depend on the unknown matrix.

Collect the source and target vectors as

$$U = [\, u_0 \mid u_2 \mid \cdots \mid u_{26} \mid e \mid w \,], \tag{12}$$

$$V = [\, -c^{(1)} \mid -c^{(3)} \mid \cdots \mid -c^{(27)} \mid v \mid e \,]. \tag{13}$$

For the concrete trajectory,

$$\det(U) = 1618594280 \neq 0 \pmod{p}. \tag{14}$$

Thus the prescribed images determine a unique linear layer,

$$M = VU^{-1} \pmod{p}. \tag{15}$$

More generally, a collection of prescribed images $MU = V$ is consistent if and only if every linear relation among the source columns is also a relation among the target columns:

$$\ker U \subseteq \ker V. \tag{16}$$

4 A Concrete 28-Round Collision

We instantiate the construction over the KoalaBear field. For the concrete instance, take

$$a = 86231345, \qquad v_0 = 1498781608. \tag{17}$$

Choose the passive coordinates of v according to (9), and set $w = a^2 v^{\circ 3}$. The complete vectors v and w are given in Appendix A.

The choice above is one member of a larger family. Once a is fixed, cubing is bijective on $\mathbb{F}_p$, so entry into the partial-round fixed point reduces to

$$b^3 + 3\big((c^{(2)})_1\big)^2 b = -a.$$

Choosing a root determines the tail; the closing condition fixes the input amplitude, and the first pair imposes only a square-root condition. The parameter v_0 does not enter these scalar equations and remains free at this stage. Thus, for any such scalar solution and choice of v_0 with nonsingular U, the interpolation $M = VU^{-1}$ realizes the collision trajectory. MDS and the Initiative's matrix checks are additional filters on this family.

For the concrete values above, the ten partial-round blocks preserve the amplitude $-a$, and the four nontrivial full-round transitions are summarized in Table 1.

Table 1. Scalar half-difference transitions across the four full-round blocks. Here s is the common coordinate-1 value before the first S-box, and b, b' are the input and output amplitudes.

Rounds	common coordinate s	input amplitude b	output amplitude b'
$0, 1$	1646804083	341544265	401295677
$2, 3$	1115766977	401295677	$2044475088 = -a$
$24, 25$	1320808719	$2044475088 = -a$	1659843992
$26, 27$	540590095	1659843992	1789162168

Each row of Table 1 satisfies (10). Since

$$341544265 = -1789162168 \pmod{p},$$

the final half-difference is the negative of the initial half-difference.

Let m_0 be the chosen input midpoint with $(m_0)_0 = \rho$, and define

$$b_0 = 1789162168, \qquad x^{\pm} = m_0 \mp b_0 e_1.$$

The complete vector m_0 and the two input states are given in Appendix A. The two inputs agree in every coordinate except coordinate 1 and both satisfy the required fixed prefix.

Theorem 1 (28-round compression collision). *For the matrix (15), the full Poseidon permutation satisfies*

$$\mathsf{P}_M(x^+) = b_0 e_1, \qquad \mathsf{P}_M(x^-) = -b_0 e_1.$$

Consequently,

$$\mathsf{H}_M(x^+) = \mathsf{H}_M(x^-) = m_0$$

in all sixteen output coordinates.

Proof. The reset equations (11) make the midpoint zero after every two-round block, so the final midpoint is zero. The first two full-round blocks send the initial half-difference $-b_0 e_1 = 341544265\, e_1$ to $-a e_1$. The ten partial-round blocks preserve $-a e_1$ by Section 3.2. The final two full-round blocks send it to $b_0 e_1$. Therefore the two permutation outputs are $\pm b_0 e_1$. Using (3),

$$\mathsf{H}_M(x^{\pm}) = \mathsf{P}_M(x^{\pm}) + x^{\pm} = \pm b_0 e_1 + (m_0 \mp b_0 e_1) = m_0.$$

5 Validation

The matrix M of (15) is MDS: every square submatrix is nonsingular over $\mathbb{F}_p$. For a 16×16 matrix this amounts to

$$\sum_{k=1}^{16} \binom{16}{k}^2 = \binom{32}{16} - 1 = 601080389$$

minors, which we verified to be nonzero with two independent exact finite-field implementations, using Dodgson (Desnanot–Jacobi) condensation and Laplace expansion.

Separately, M clears the Initiative's linear-layer security gate against infinitely long subspace trails [8]: the reference verifier at the pinned commit of [4] returns `verify_mds_matrix(M,p)=True`. Both facts, and the collision itself, are reproducible against that implementation; in particular its collision verifier returns `True` at $t = 16$, i.e. in all sixteen output coordinates. The exact matrix, construction parameters, and collision are listed in Appendix A.

6 Discussion

Adaptive parameter selection. The construction uses relations of the form

$$Mu_r = -c^{(r+1)}$$

between the selected matrix and the already fixed round constants. MDS tests constrain diffusion through the minors of M, while the subspace-trail tests of [8] constrain persistent subspace behavior of M and its powers. Neither class of test constrains these joint matrix–constant relations. The attack therefore identifies the parameter-selection order as an additional part of the security analysis.

A standard way to avoid such adaptive correlations is to derive the complete parameter set non-adaptively, for example from a common seed or by binding the round constants to a canonical encoding of the matrix. Then equations such as $Mu_r = -c^{(r+1)}$ are simultaneous constraints on jointly determined parameters rather than freely programmable linear images.

Image constraints. For a direct construction spanning L two-round blocks, midpoint reset uses L reset images together with two reusable difference images. If all source vectors are independent, this requires

$$L + 2 \leq t.$$

The present instance has $L = 14$ and $t = 16$, so the direct construction uses a basis of the state space. This count is not a general limitation: by (16), additional prescribed images are possible whenever the linear dependencies among their sources and targets are compatible.

Relation to recent attacks. The collision reported here was submitted to the Poseidon Cryptanalysis Initiative on 31 July 2026, before the public appearance of Slipway [10] and the attack of Bhati, Tariq, and Ashur [3]. The works were developed independently.

Slipway is close in its parameter-selection viewpoint: it also constructs a round-constant-dependent MDS matrix and exploits finite subspace trails through the partial rounds. Its mechanism steers a low-dimensional family into such a trail. Midpoint reset instead follows two executions and programs an exact two-round trajectory whose difference is recycled and whose midpoint is repeatedly canceled.

The attack of Bhati, Tariq, and Ashur takes a complementary direction. It keeps the linear layer fixed and restricts selected partial-round S-box inputs so that rounds can be skipped in the algebraic description. Midpoint reset uses the opposite degree of freedom: it fixes the two-execution trajectory first and then chooses the linear layer to realize the required images.

7 Conclusion

We presented a full 28-round compression collision for the width-16 KoalaBear Poseidon instance when the MDS matrix may be chosen after the round constants are fixed. The construction repeatedly resets the midpoint of two executions and reuses a short difference cycle to carry their half-difference through both full and partial rounds. These trajectory constraints determine an MDS linear layer that passes the prescribed subspace-trail checks, while the final difference cancels under feed-forward. The result shows that security analysis for generated Poseidon parameters must account not only for the properties of the individual components but also for how those components are selected together.

Acknowledgments. We thank the Ethereum Foundation Poseidon Group for organizing the Poseidon Cryptanalysis Initiative and for reviewing the submitted collision.

AI usage disclosure. ChatGPT Pro was used during the development of the main idea presented in this paper. OpenAI Codex assisted with manuscript editing and implementation of computational checks. The author independently verified the mathematical arguments, references, and computational results.

References

1. Bak, A., Bariant, A., Boeuf, A., Hostettler, M., Jazeron, G.: Claiming bounties on small scale Poseidon and Poseidon2 instances using resultant-based algebraic attacks. Cryptology ePrint Archive, Paper 2026/150 (2026)
2. Bariant, A., Bouvier, C., Leurent, G., Perrin, L.: Algebraic attacks against some arithmetization-oriented primitives. IACR Transactions on Symmetric Cryptology **2022**(3), 73–101 (2022). https://doi.org/10.46586/tosc.v2022.i3.73-101
3. Bhati, A.S., Tariq, S., Ashur, T.: From Round Skipping to S-Box Skipping: Attacking Poseidon's Partial Layer via Subspace Restriction. Cryptology ePrint Archive, Paper 2026/1692 (2026)
4. Ethereum Foundation Poseidon Group: Poseidon Cryptanalysis Initiative 2024–2026. Reference implementation https://github.com/khovratovich/poseidon-tools, commit 60075da7c0521d9493749a035b1f30d4eda37138; accessed 31 July 2026
5. Grassi, L., Khovratovich, D., Rechberger, C., Roy, A., Schofnegger, M.: Poseidon: A New Hash Function for Zero-Knowledge Proof Systems. In: 30th USENIX Security Symposium (USENIX Security 21), pp. 519–535. USENIX Association (2021)

6. Grassi, L., Koschatko, K., Rechberger, C.: Poseidon and Neptune: Gröbner Basis Cryptanalysis Exploiting Subspace Trails. IACR Transactions on Symmetric Cryptology **2025**(2), 34–86 (2025). https://doi.org/10.46586/tosc.v2025.i2.34-86
7. Grassi, L., Lüftenegger, R., Rechberger, C., Rotaru, D., Schofnegger, M.: On a Generalization of Substitution-Permutation Networks: The HADES Design Strategy. In: Advances in Cryptology – EUROCRYPT 2020, Part II. Lecture Notes in Computer Science, vol. 12106, pp. 674–704. Springer (2020)
8. Grassi, L., Rechberger, C., Schofnegger, M.: Proving Resistance Against Infinitely Long Subspace Trails: How to Choose the Linear Layer. IACR Transactions on Symmetric Cryptology **2021**(2), 314–352 (2021). https://doi.org/10.46586/tosc.v2021.i2.314-352
9. Sanso, A., Vitto, G.: Top Gun: Degree Annihilation Attacks on Poseidon. Cryptology ePrint Archive, Paper 2026/1254 (2026)
10. Vitto, G.: Slipway: Accessing Finite Subspace Trails in Poseidon. Cryptology ePrint Archive, Paper 2026/1579 (2026)

A Concrete Matrix, Parameters, and Collision

Write $M = (M_L \mid M_R)$, where $M_L, M_R \in \mathbb{F}_p^{16\times 8}$. All entries below are represented by their canonical integers in $\{0, \ldots, p-1\}$.

$$M_L = \begin{pmatrix}
1707690736 & 1498781608 & 1419077821 & 2194016615 & 5262857413 & 3532598 & 4 & 868711018 \\
5996986 & 2064317645 & 1969501547 & 1567065873 & 1734425278 & 1527898552 & 1225220284 & 1303967628 \\
1062823282 & 1105518764 & 161204886 & 1115509429 & 774851300 & 1377548372 & 642042297 & 943084981 \\
644690960 & 1897941121 & 1969501547 & 560717268 & 1016752661 & 1401475731 & 3360867981 & 1102168 \\
57 & 215841447 & 425758093 & 161204886 & 1549673840 & 659340453 & 3805602701 & 7703142681 \\
4067267381 & 466142735 & 1563330936 & 1969501547 & 1345360076 & 257572053 & 531991303 & 93563767 \\
123926721 & 155428770 & 384299387 & 161204886 & 1006771014 & 3066869321 & 1648865301 & 982088547 \\
7225951621 & 874892582 & 870113079 & 1969501547 & 1391869094 & 1693192160 & 1954545710 & 7976540251 \\
7933987611 & 470568472 & 843046868 & 161204886 & 1350811500 & 2036148147 & 614166672 & 937422807 \\
5975469941 & 234381567 & 38439290 & 1969501547 & 1201537456 & 1533365856 & 865415549 & 8602837391 \\
300775256 & 145019581 & 2087270745 & 1969501547 & 284154262 & 200082514 & 543373627 & 154185830 \\
7905490761 & 115224401 & 2086706889 & 1969501547 & 2046177012 & 971883392 & 643124781 & 2273217212 \\
005851776 & 402913336 & 847558908 & 1969501547 & 1696673004 & 286619019 & 5260320051 & 7144018861 \\
527172951 & 062843232 & 1688217959 & 161204886 & 1870860577 & 2184344861 & 4531501611 & 4969634991 \\
682331939 & 860436737 & 26959235 & 161204886 & 1713421211 & 2127859359 & 635765663 & 4904073371 \\
237642129 & 1311044048 & 1780487358 & 161204886 & 1061895537 & 1473276095 & 1254839743 & 1716277124 \\
1065903629 & 1849600089
\end{pmatrix},$$

$$M_R = \begin{pmatrix}
683782132 & 979359873 & 1926020330 & 208238739 & 2336813011 & 4491007991 & 280029501 & 174917492 \\
940890632 & 1384355174 & 471540849 & 1322928054 & 778512346 & 766647959 & 3873102021 & 571127024 \\
1404116300 & 974945006 & 1948947066 & 1451260530 & 2520660581 & 209126029 & 559887889 & 19665855 \\
679129778 & 906030353 & 724608762 & 669503878 & 1308233201 & 298798821 & 863600910 & 341845639 \\
540220758 & 487234058 & 1816116464 & 1146735801 & 751619707 & 374132328 & 3918722151 & 424260832 \\
7536986371 & 699622065 & 1531723364 & 168415643 & 665673552 & 9688657662 & 117566398 & 892444570 \\
1778074094 & 242818853 & 1088178535 & 1907486926 & 700812581 & 797406448 & 1968691624 & 1966577122 \\
154616574 & 1507302768 & 279648995 & 2013902545 & 2653333022 & 016564690 & 231133083 & 91935800 \\
1692155928 & 1329772876 & 1403450702 & 561591060 & 1565292476 & 1265325234 & 1213963076 & 217881738 \\
1657245249 & 1214019205 & 147149775 & 1095481651 & 794284547 & 609277980 & 168249425 & 274560838 \\
1813730144 & 974098967 & 755784963 & 1365552788 & 807496242 & 81014680 & 2103716441 & 536134835 \\
1948678612 & 1062481989 & 506214724 & 1619714855 & 982271002 & 1227059424 & 1306081916 & 55200809 \\
1982127754 & 210960858 & 147519874 & 1668372657 & 631656843 & 1344581331 & 1329580230 & 746294950 \\
310031353 & 1186551204 & 69410778 & 524359642 & 1322816821 & 2124709351 & 2096100759 & 26321414 \\
7340339692 & 093326358 & 190554092 & 750477735 & 419477229 & 978596588 & 489651895 & 289478493 \\
1112904393 & 1903742495 & 812278242 & 205936272 & 702043870 & 2753787801 & 4264289861 & 332569841
\end{pmatrix}.$$

The difference-cycle parameter is

$$a = 86231345,$$

and the vectors v and $w = a^2 v^{\circ 3}$ are

$$\begin{aligned}
v = (&1498781608, 1969501547, 161204886, 1969501547,\\
&161204886, 1969501547, 161204886, 1969501547,\\
&161204886, 1969501547, 1969501547, 1969501547,\\
&1969501547, 161204886, 161204886, 161204886)^T,\\
w = (&331772206, 1969501547, 161204886, 1969501547,\\
&161204886, 1969501547, 161204886, 1969501547,\\
&161204886, 1969501547, 1969501547, 1969501547,\\
&1969501547, 161204886, 161204886, 161204886)^T.
\end{aligned}$$

The input midpoint used in Section 4 is

$$\begin{aligned}
m_0 = (&12623332, 381610402, 1098917770, 1214689931,\\
&1088609520, 1417244378, 389562611, 528440087,\\
&1037009343, 103050012, 1856868908, 844069,\\
&2047831014, 2081742232, 501746317, 1636319921)^T.
\end{aligned}$$

With $b_0 = 1789162168$, the two complete input states $x^+ = m_0 - b_0 e_1$ and $x^- = m_0 + b_0 e_1$ are

$$\begin{aligned} x^+ = (&12623332, 723154667, 1098917770, 1214689931,\\ &1088609520, 1417244378, 389562611, 528440087,\\ &1037009343, 103050012, 1856868908, 844069,\\ &2047831014, 2081742232, 501746317, 1636319921)^T,\\ x^- = (&12623332, 40066137, 1098917770, 1214689931,\\ &1088609520, 1417244378, 389562611, 528440087,\\ &1037009343, 103050012, 1856868908, 844069,\\ &2047831014, 2081742232, 501746317, 1636319921)^T. \end{aligned}$$

Direct evaluation of all 28 rounds gives

$$\begin{aligned} \mathsf{P}_M(x^+) &= (0, 1789162168, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)^T,\\ \mathsf{P}_M(x^-) &= (0, 341544265, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)^T. \end{aligned}$$

Consequently,

$$\mathsf{H}_M(x^+) = \mathsf{H}_M(x^-) = m_0.$$